

ALGORITMI OBDELAVE PODATKOV DALJINSKEGA ZAZNAVANJA

DOMEN MONGUS

Vsebina

- Diferencialni morfološki profili
- Teksturna analiza
 - ▣ Histogrami (hitra ponovitve)
 - ▣ Sivinska matrika sopojavitev

Diferencialni profil

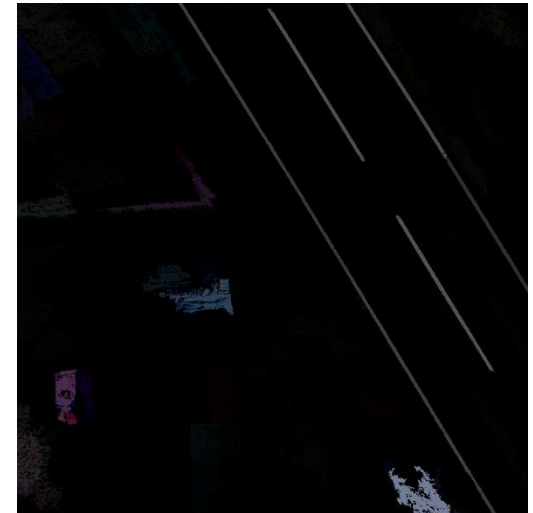
- Profiliranje z zaporednimi cilindričnimi transformacijami:
 - ▣ Odstranjene strukture, ki sovpadajo s filtri



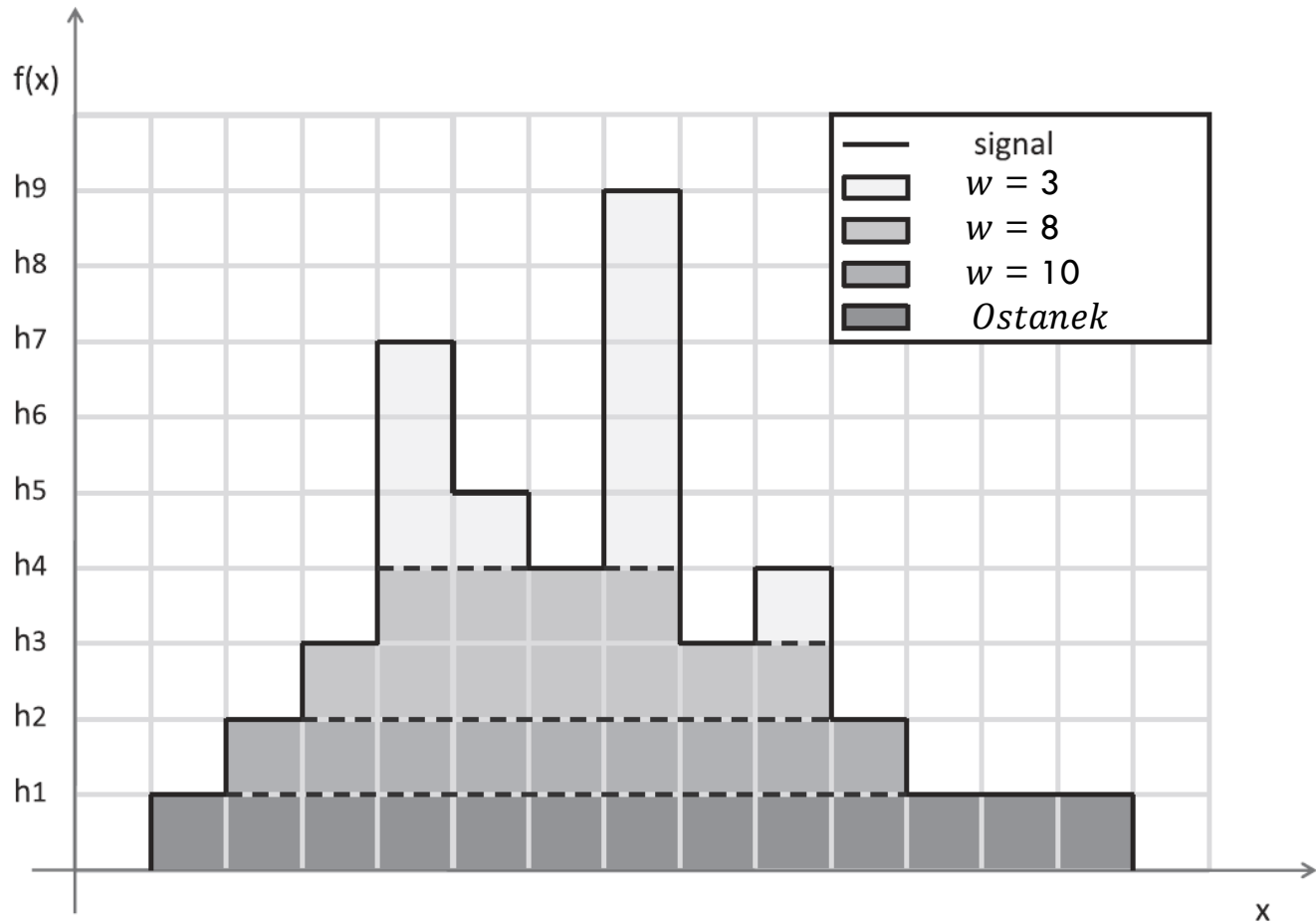
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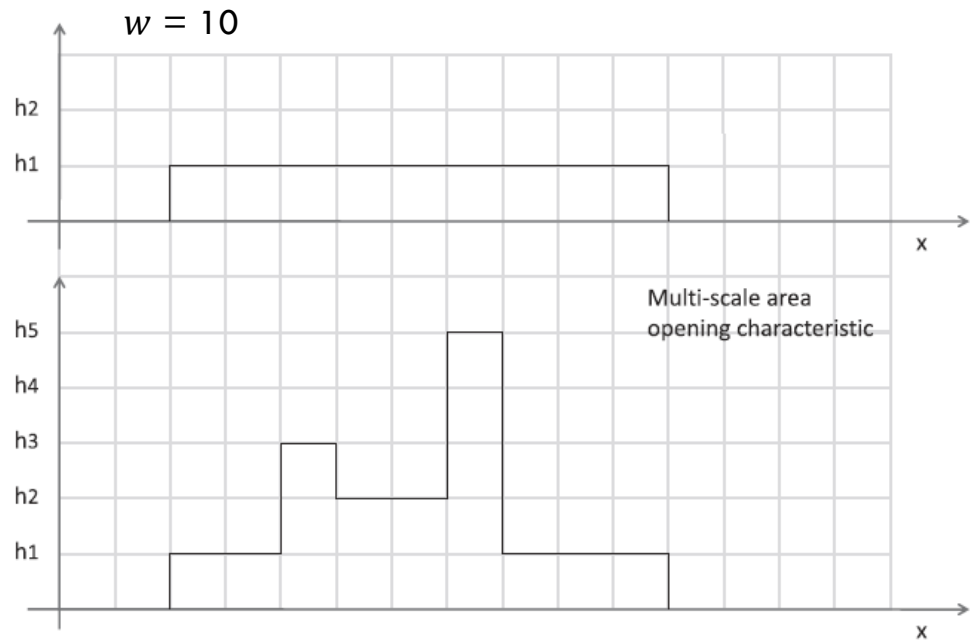
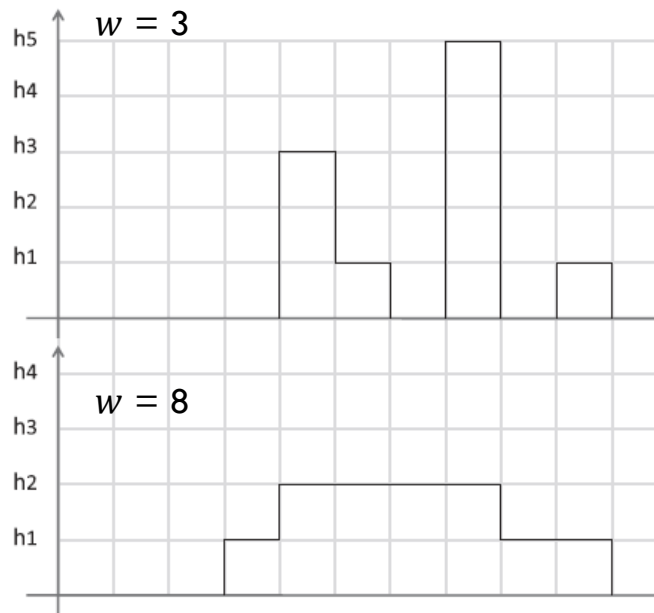


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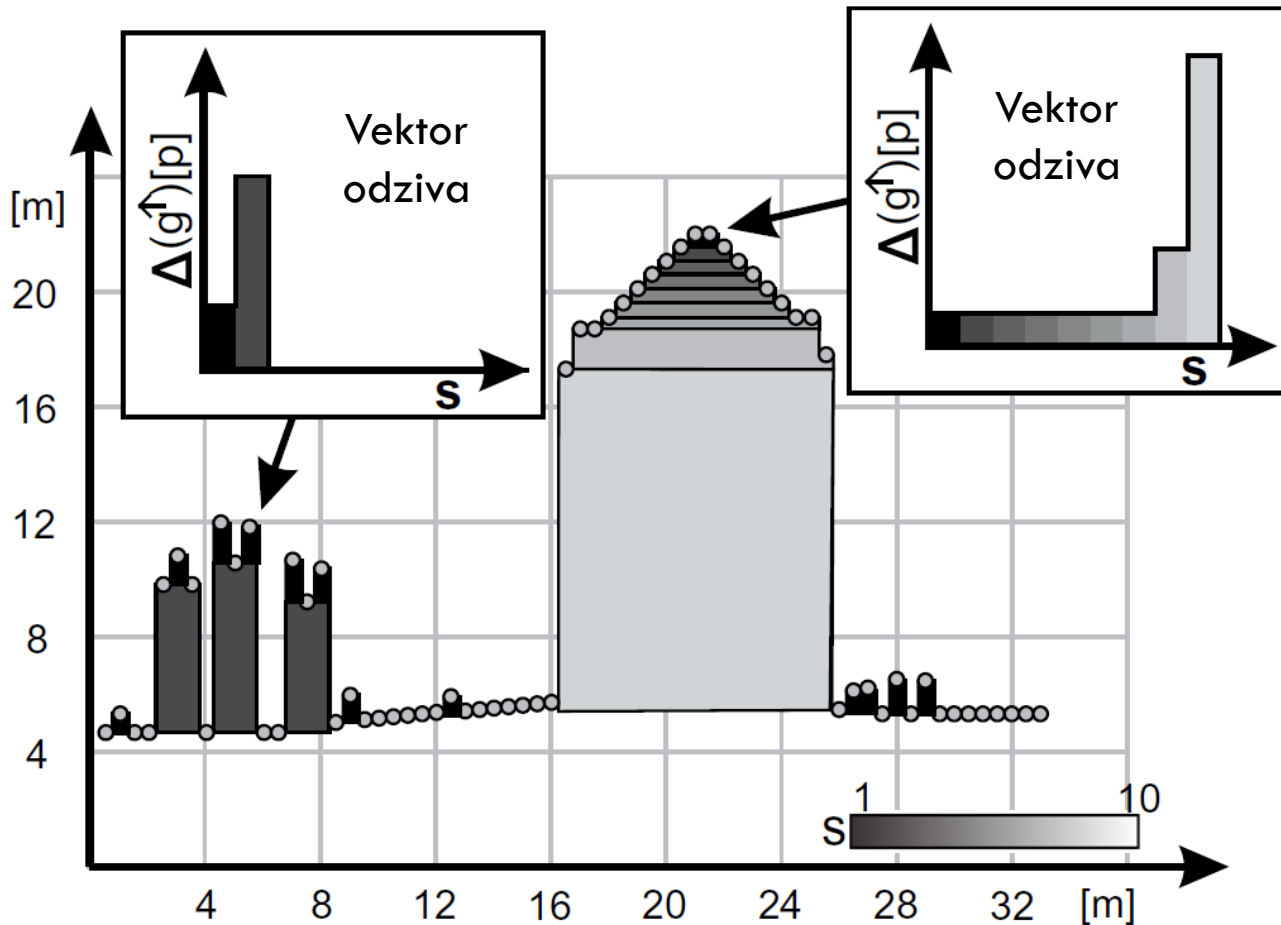


Morfološka dekompozicija

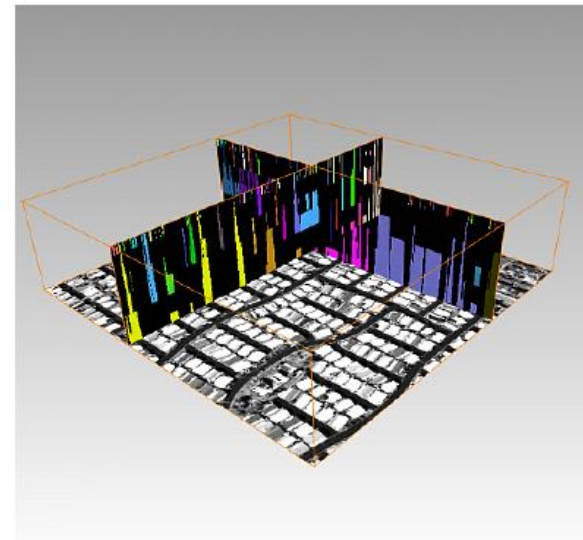
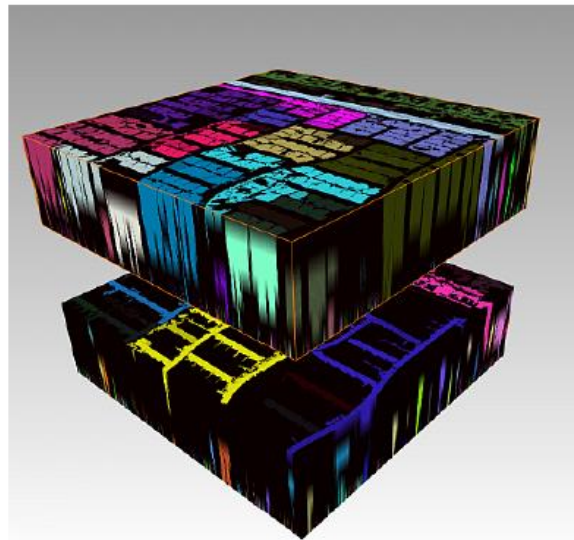
□ Diferencialni morfološki profili:



Morfološka dekompozicija

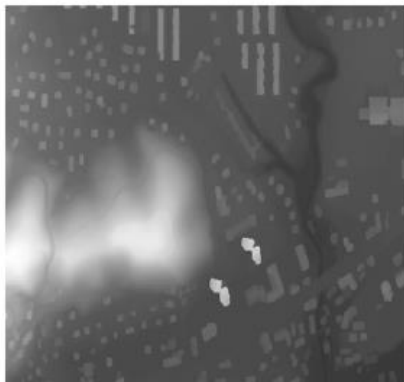


Morfološka dekompozicija



Mapiranje karakterističnih vrednosti

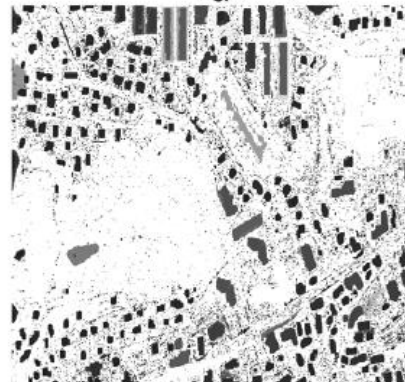
Vhod



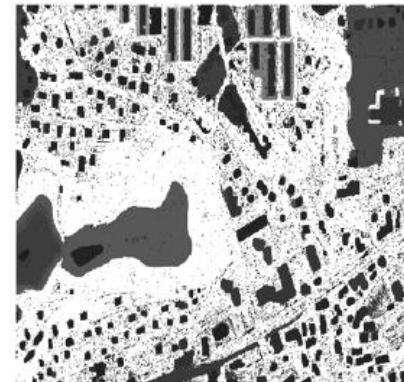
Največji odzivi



Velikost okna

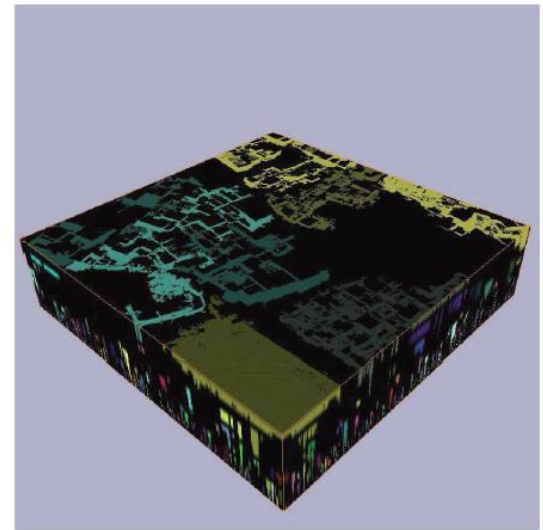
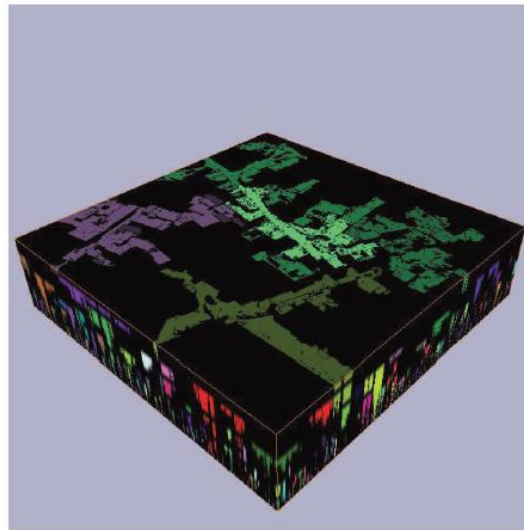


Deviacija odzivov



Hribi in doline

- Po enakem principu izdelamo profile za zapiranje



- Primerjamo profile:
 - ▣ Če je večji dobljen z odpiranjem je hrib, sicer dolina

Analiza teksture

- Osnovni pristopi:

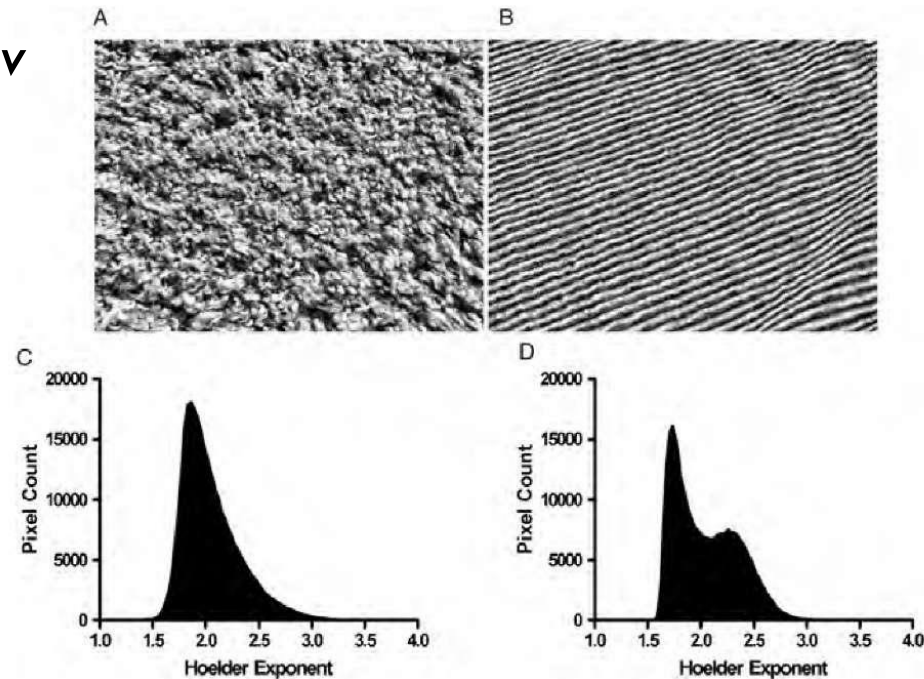
- ▣ Histogrami

- ▣ Sivinska sopojavitvena matrika

- Histogram $H[k] = \text{število elementov v košu } k$

- ▣ $K = \text{število košev}$

- ▣ $k \in [0, K-1]$



Analiza teksture - Histogram

□ Kako določiti število košev K ?

□ $K = \frac{Max\ n - Min\ n}{h}$, kjer je h velikost koša

□ $K = \sqrt{n}$

□ Normalna porazdelitev običajno

$$k = \lceil \log_2 n \rceil + 1,$$

□ Izboljšava za nenormalno porazdelitev

$$k = 1 + \log_2(n) + \log_2 \left(1 + \frac{|g_1|}{\sigma_{g_1}} \right)$$

Analiza teksture - Histogram

- Povprečje na osnovi histograma

$$\bar{n} = \frac{\sum_{k=0}^K k * H[k]}{\sum_{k=0}^K H[k]}$$

- Pričakovana vrednost = $\arg \max H[k]$

- Kako izračunamo mediano?

- Standardni odklon

$$\bar{n} = \sqrt{\frac{\sum_{k=0}^K k * (H[k] - \bar{n})}{\sum_{k=0}^K H[k]}}$$

Analiza tekstone - Histogram

□ Spremenljivost

$$m_1 = \frac{\sum (x_i - \bar{x})}{N}$$

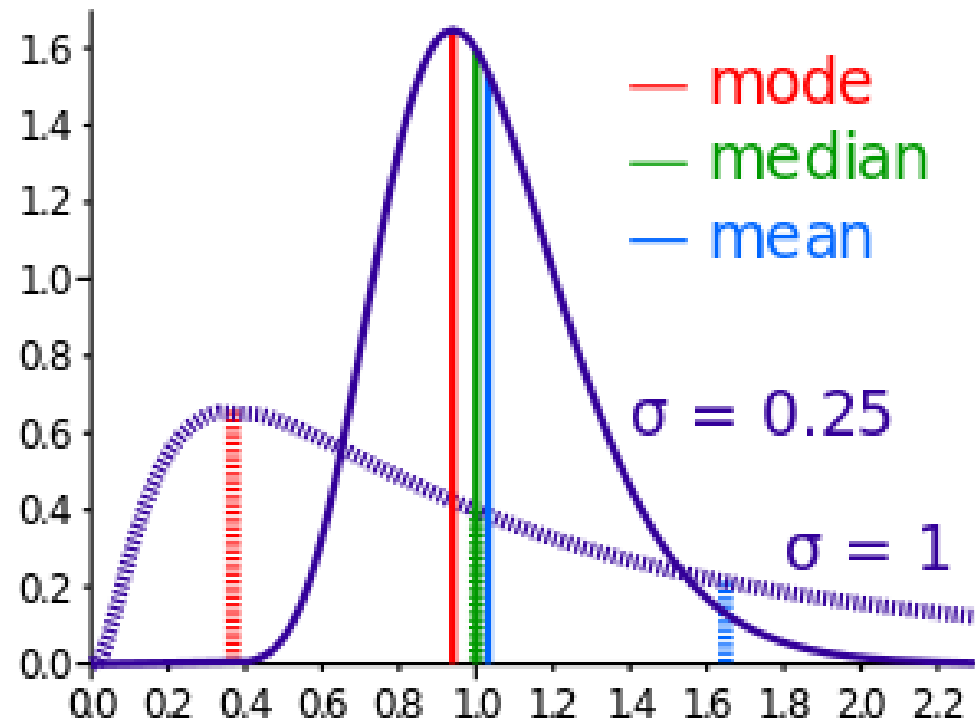
$$m_2 = \frac{\sum (x_i - \bar{x})^2}{N}$$

$$m_3 = \frac{\sum (x_i - \bar{x})^3}{N}$$

$$m_4 = \frac{\sum (x_i - \bar{x})^4}{N}$$

Asimetrija (skewness)

$$b_1 = \frac{m_3}{s^3} = \frac{\frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^3}{\left[\frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2 \right]^{3/2}},$$



Analiza tekstone - Histogram

□ Spremenljivost

$$m_1 = \frac{\sum (x_i - \bar{x})}{N}$$

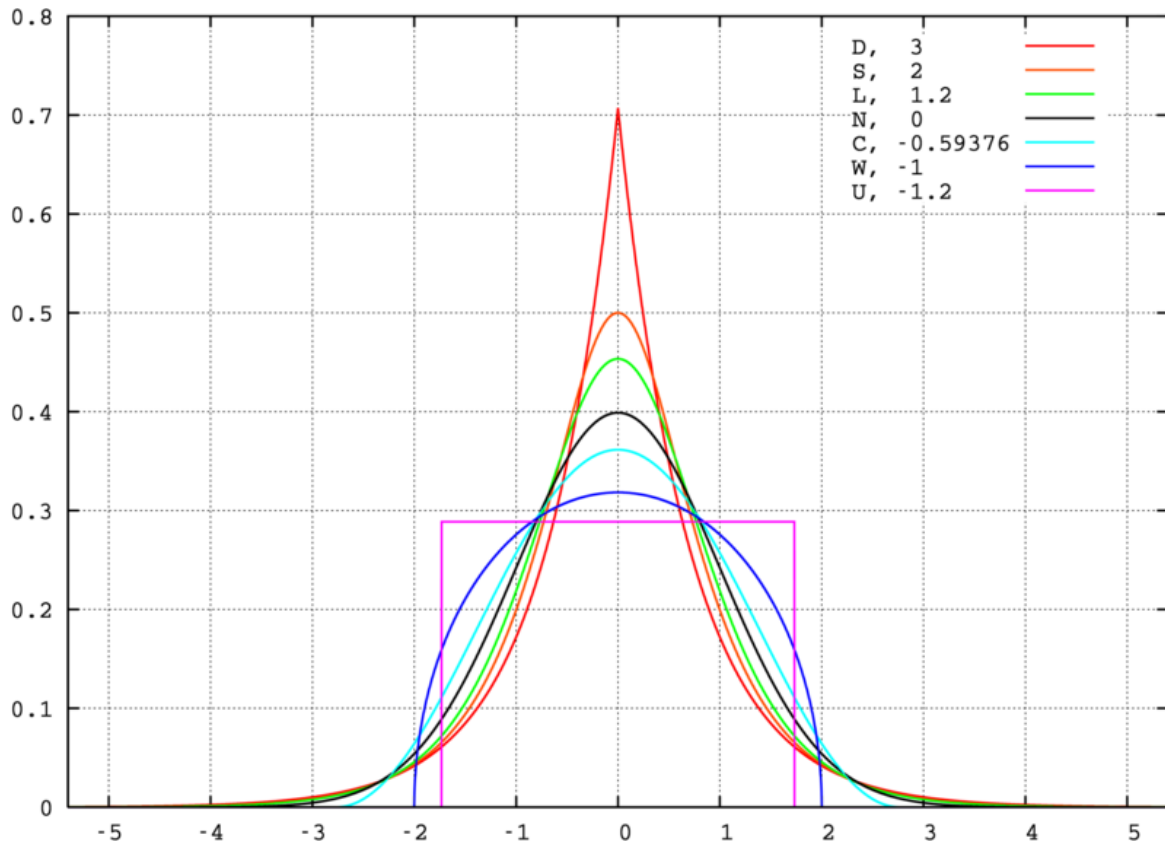
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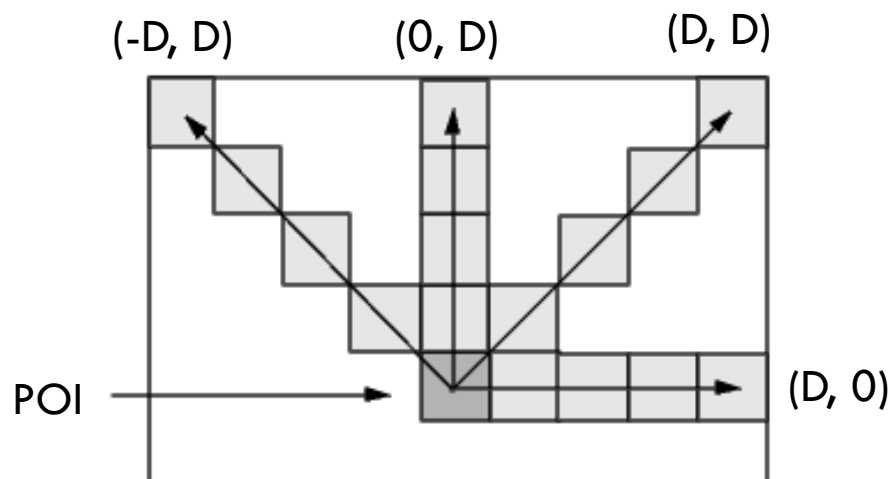
Sploščenost (kurtosis)

$$g_2 = \frac{m_4}{m_2^2} - 3 = \frac{\frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^4}{\left(\frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^2\right)^2} - 3$$



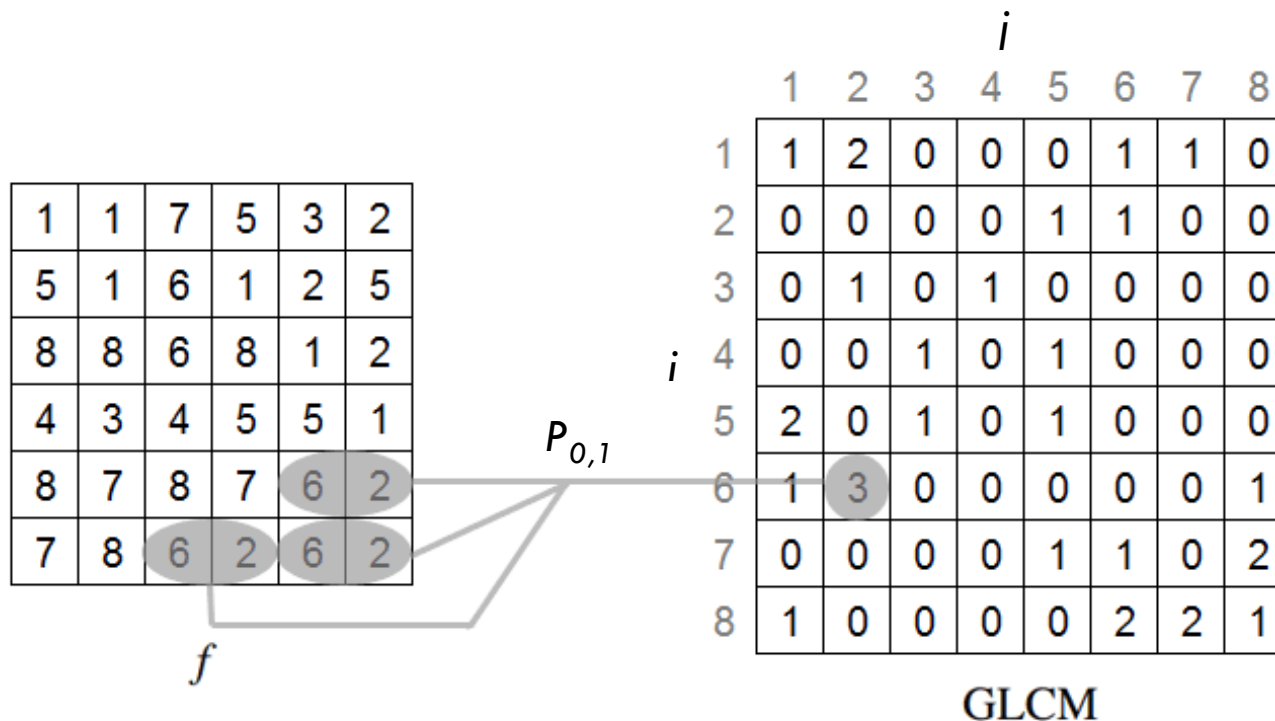
Matrika sopoljavitve sivinskih vrednosti

- Ang. gray level cooccurrence matrix
 - ▣ GLCM
- Porazidelitev parov vrednosti na določenem odmiku
- GLMC podaja frekvenco pojavitve piksla z vrednostjo i na podanem odmiku (horizontalno, vertikalno ali diagonalno) od vrednosti j
 - ▣ Notacija odmikov: $P_{0,D}$, $P_{D,0}$, $P_{D,D}$, $P_{-D,D}$



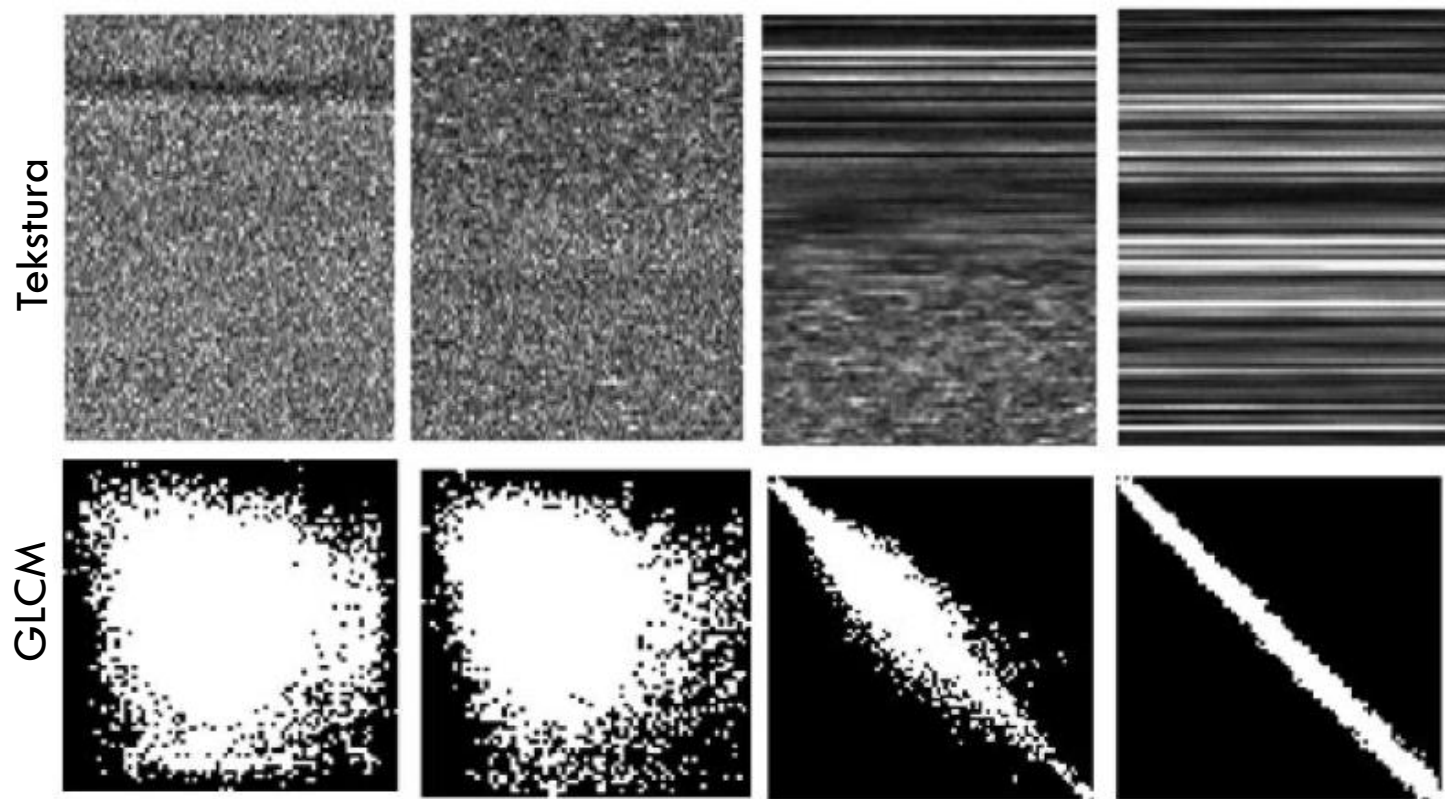
Matrika so pojavitve sivinskih vrednosti

- GLCM je v bistvu histogram so pojavitvenih vrednosti



Matrika sopojavitve sivinskih vrednosti

- GLCM je v bistvu histogram sopojavitvenih vrednosti



Statistike nad GLCM

- Homogenost

$$H(P) = \sum_{i=0}^{255} \sum_{j=0}^{255} \frac{GLCM(i, j)}{1 + (i - j)^2}$$

- Kontrast

$$C(P) = \sum_{i=0}^{255} \sum_{j=0}^{255} GLCM(i, j)(i - j)^2$$

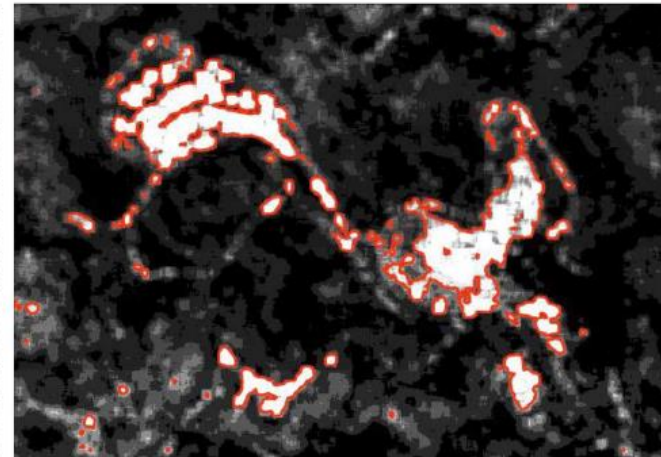
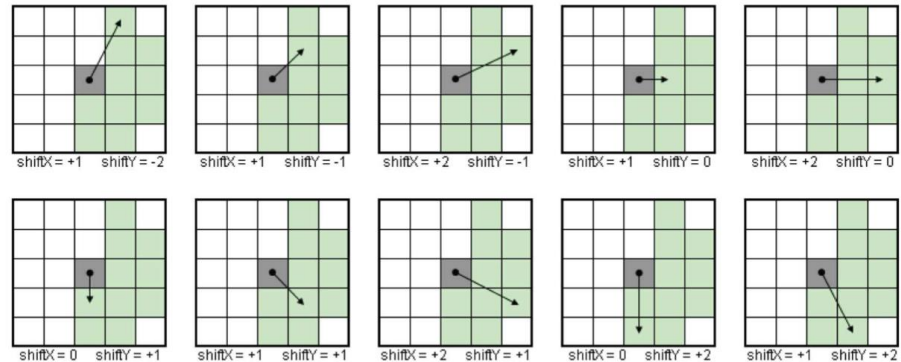
- Entropija

$$E(P) = \sum_{i=0}^{255} \sum_{j=0}^{255} -GLCM(i, j) \log_e(GLCM(i, j))$$

Primer uporabe

- ❑ PANTEX v uporabi EC za detekcijo naselij (predvsem begunskih kampov in nelegalnih naselji)
- ❑ Definicija

- $\text{Min}(C(P))$, kjer $P =$



Primer uporabe

□ Jeruzalem:

